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Sciadopitysin mitigates spermatological and testicular damage instigated by paraquat administration in male albino rats

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Paraquat (PQ) is a herbicide that has ability to induce testicular toxicity by producing reactive oxygen species (ROS). Sciadopitysin (SPS) is a promising flavonoid that displays multiple pharmacological properties i.e., anti-inflammatory, anti-oxidant and anti-apoptotic. Therefore, the present study was designed to evaluate the mitigative role of SPS against PQ induced testicular toxicity in male rats. The experiment was performed on male albino rats ($n = 40$), that were divided into 4 groups. The group-1 was control group. Group-2 was administrated orally with PQ (5 mg/kg). Group-3 was administrated orally with PQ (5 mg/kg) and SPS (2 mg/kg). Group-4 was supplemented with SPS (2 mg/kg) through oral gavage. The experiment was conducted for 56 days. The exposure to PQ significantly lowered the activities of catalase (CAT), glutathione reductase (GSR), superoxide dismutase (SOD) as well as glutathione peroxidase (GPx). Whereas, a substantial increase was observed in dead sperms number, abnormalities in the tail, head as well as midpiece of sperms in PQ intoxicated rats. Moreover, a significant increase in the level of ROS and malondialdehyde (MDA) was noticed in PQ administrated group. Furthermore, steroidogenic enzymes expression was significantly decreased in PQ-intoxicated group, whereas the level of inflammatory markers was increased in PQ administrated rats. Besides, the expression of apoptotic markers was significantly escalated in PQ exposed rats, whereas the expression of anti-apoptotic markers was considerably reduced. A significant reduction in hormonal level was also noticed in the rats that were administrated with PQ. Moreover, the histopathological examination revealed that PQ significantly damaged the testicles. However, the supplementation of SPS with PQ significantly reduced the adverse effects of PQ in the testes of albino rats. Therefore, the current investigation demonstrated that SPS possesses a significant potential to avert PQ-induced testicular dysfunction due to its anti-apoptotic, androgenic, anti-oxidant and anti-inflammatory properties.

Herbicides are toxic chemical compounds that are used to control unwanted plants as well as weeds¹. PQ is an organic chemical that is widely applied as an herbicide^{2,3} to control weeds in multiple crops such as tea, rice, maize crops and sugar cane⁴. The toxicity of PQ is considered as a critical issue in developing countries especially in Asia. However, PQ acute intoxication induces organ injury with a death ratio of more than 60–70%⁵. Human exposure to PQ occurs through inhalation, skin and ingestion; its profound exposure may cause death within 3.5 hours⁶. PQ adversely affects the different parts of the body by increasing pro-inflammatory markers, i.e., IL-6, decreasing dopamine and accumulating Lewy bodies⁷.

Previous literature has revealed that PQ exposure leads to pulmonary toxicity, nephrotoxicity, hepatotoxicity, brain damage¹ and reproductive toxicity such as testicular dysfunction⁸. The major Oxidative stress (OS) and ROS are the culprits of PQ-induced toxicity. Physiological level of ROS usually regulates the inflammatory reactions, growth factors as well as cell signaling. On the other hand, excessive ROS generation leads to severe damages⁹. ROS affects sperm quality by damaging the sperm plasma membrane, as the plasma membrane comprises of

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high concentration of polyunsaturated fatty acids (PUFAs) and prompts sperm structural abnormalities by inducing lipid peroxidation¹⁰. It is reported that PQ exposure also results in reduced spermatocyte, spermatids as well as Leydig cells number¹¹.

Polyphenolic flavonoids and biflavonoids are getting attention due to their unique pharmacological potentials as they are used against various disorders^{12,13}. Biflavonoids are naturally reported in gymnosperms and pteridophytes that have flavonoid dimers attached to each other in their structure. SPS is an important amentoflavone derivative biflavonoids naturally present in *Taxus chinensis* that shows anti-inflammatory and anti-oxidative properties¹⁴. According to previous literature, SPS inhibits receptor activator of nuclear factor-kappa B ligand-induced osteoclastogenesis and bone loss by inhibiting NF- κ B activation¹⁵. However, the mitigative role of SPS in attenuating the testicular toxicity has not yet been reported. Therefore, the current study was planned to evaluate the mitigative role of SPS against PQ induced testicular damage in male albino rats.

Results

Effect of PQ and SPS on biochemical markers

PQ exposure significantly ($P < 0.01$) decreased the activities of antioxidant enzymes i.e., CAT, GPx, SOD and GSR, on the other hand increased ROS and MDA levels in PQ administrated group as compared to the control group. However, the administration of SPS with PQ resulted in a significant ($P < 0.05$) increase in CAT, GPx, SOD and GSR activities, whereas, a significant reduction was noticed in ROS and MDA levels as compared to PQ exposed group. Moreover, only SPS supplemented group displayed the similar values as to the control group (Table 1).

Effect of PQ and SPS on semen profile

The administration of PQ significantly ($P < 0.01$) reduced sperm viability, motility and number whereas, increased the sperm structural abnormalities (tail, mid-piece and head) in contrast to control rats. However, SPS administration significantly ($P < 0.001$) reversed all the damages in co-treated rats as compared to PQ administrated rats. Moreover, only SPS supplemented group showed semen profile similar to the control group (Table 2).

Effect of PQ and SPS on apoptotic markers

The exposure to PQ significantly ($P < 0.05$) increased Bax and Caspase-3 expressions, while reducing the expression of Bcl-2 in PQ exposed group as compared to control group. On the other hand, SPS supplementation significantly ($P < 0.01$) reduced Bax and Caspase-3 expressions and increased the Bcl-2 expression in PQ + SPS co-treated rats in comparison to PQ administrated rats. Moreover, these expressions in SPS alone administrated rats were comparable to control rats (Fig. 1).

Parameters	Groups			
	Control	PQ	PQ + SPS	SPS
CAT (Umg ⁻¹ protein)	10.60 ± 0.33 ^a	4.86 ± 0.22 ^c	8.12 ± 0.12 ^b	10.66 ± 0.35 ^a
SOD (Umg ⁻¹ protein)	7.05 ± 0.08 ^a	2.91 ± 0.13 ^c	5.44 ± 0.27 ^b	7.11 ± 0.11 ^a
GPx (Umg ⁻¹ protein)	27.74 ± 1.02 ^a	9.81 ± 0.16 ^c	19.22 ± 0.69 ^b	28.31 ± 1.19 ^a
GSR (nM NADPH oxid/min/mg tissue)	5.84 ± 0.14 ^a	1.34 ± 0.08 ^c	4.59 ± 0.17 ^b	5.89 ± 0.17 ^a
ROS (Umg ⁻¹ protein)	0.58 ± 0.07 ^a	6.77 ± 0.11 ^c	1.89 ± 0.14 ^b	0.56 ± 0.08 ^a
MDA (nmol/mg protein)	0.73 ± 0.06 ^a	3.23 ± 0.12 ^c	1.38 ± 0.09 ^b	0.71 ± 0.06 ^a

Table 1. Mean ± SE of biochemical markers in control, PQ administrated, co-administrated and SPS administrated groups. Values with dissimilar letters are substantially ($P < 0.05$) dissimilar from other groups. PQ: Paraquat; SPS Sciadopitysin; CAT: Catalase; SOD: Superoxide dismutase; GPx: Glutathione peroxidase; GSR: Glutathione reductase; MDA: Malondialdehyde; ROS: reactive oxygen species.

Parameters	Groups			
	Control	PQ	PQ + SPS	SPS
Epididymal Sperm Count (million/mL)	25.43 ± 1.22 ^a	7.27 ± 0.45 ^c	17.06 ± 0.68 ^b	25.84 ± 1.21 ^a
Motility (%)	83.98 ± 1.28 ^a	31.56 ± 1.48 ^c	71.36 ± 2.06 ^b	84.44 ± 1.45 ^a
Dead sperms (%)	8.06 ± 1.06 ^a	75.48 ± 2.45 ^c	25.28 ± 1.38 ^b	8.11 ± 1.08 ^a
Head abnormality (%)	3.14 ± 0.16 ^a	29.84 ± 1.10 ^c	7.57 ± 0.71 ^b	3.11 ± 0.16 ^a
Mid sperm abnormality (%)	0.75 ± 0.05 ^a	6.95 ± 0.43 ^c	2.12 ± 0.11 ^b	0.72 ± 0.04 ^a
Tail abnormality (%)	2.76 ± 0.11 ^a	19.39 ± 0.16 ^c	8.66 ± 0.29 ^b	2.73 ± 0.11 ^a
Hypo- osmotic swelled sperm count (HOS) (%)	81.70 ± 1.51 ^a	19.62 ± 1.16 ^c	63.43 ± 2.92 ^b	84.60 ± 1.77 ^a

Table 2. Mean ± SE of sperm indices in control, PQ administrated, co-administrated and SPS administrated groups. Values with dissimilar letters are notably ($P < 0.05$) dissimilar from other groups. PQ: Paraquat; SPS Sciadopitysin.

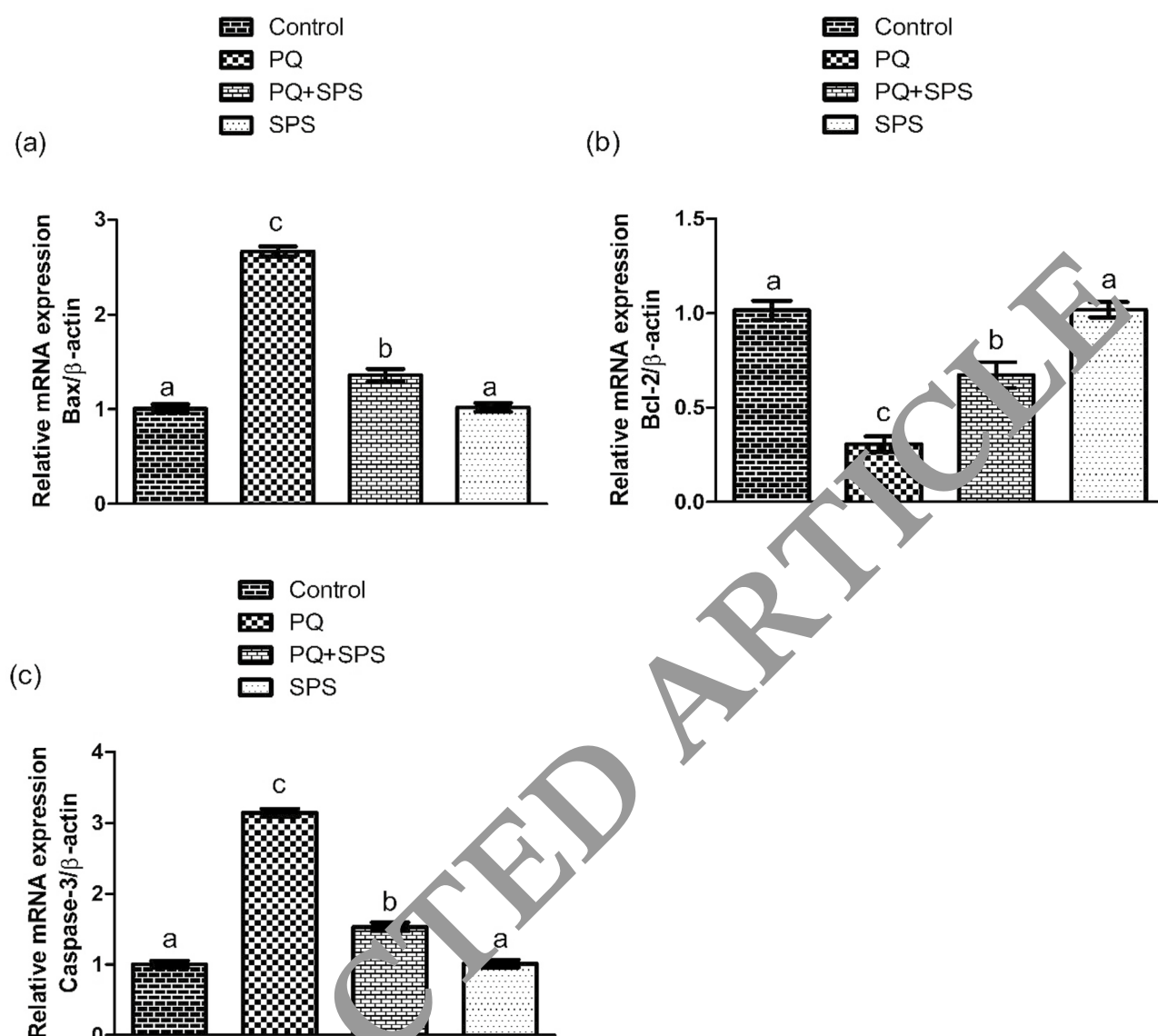


Figure 1. Effect of PQ and SPS on the expression of (a) Bax, (b) Bcl-2 and (c) caspase-3. Vertical bars are presented on mean $\pm$ SE values. Different superscripts on bars are showing significant ($P < 0.05$) difference. PQ: Paraquat; SPS: Sciadopitysin; Bcl-2: B-cell lymphoma 2; Bax: Bcl-2-associated X protein; Caspase-3: Cysteine-dependent protease.

Effect of PQ and SPS on inflammatory indices

The levels of inflammatory markers were significantly ($P < 0.01$) increased in PQ administrated group in contrast to control group. However, the co-administration of PQ + SPS significantly ($P < 0.05$) decreased the inflammatory indices as compared to PQ exposed group. Moreover, these markers in SPS alone administrated rats were similar to the control rats (Table 3).

Effect of PQ and SPS on steroidogenic enzymes

PQ intoxication resulted in a significant ($P < 0.001$) reduction in the expression of steroidogenic enzymes (3β-HSD, 17β-HSD and StAR) as compared to control group. However, the expressions of 17β-HSD, 3β-HSD and StAR in co-administrated rats (PQ + SPS) were significantly ($P < 0.01$) increased in comparison to PQ administrated group. Moreover, these expressions in SPS alone administrated group were near to the control group (Fig. 2).

Effect of PQ and SPS on hormonal assay

The administration of PQ significantly ($P < 0.01$) reduced the level of FSH, LH and testosterone as compared to control rats. However, the co-treated rats (PQ + SPS) displayed a significant ($P < 0.05$) increase in the levels of these hormones in comparison to PQ administrated rats. Furthermore, the levels of these hormones in SPS only administrated rats were comparable to the control group (Table 4).

Parameters	Groups			
	Control	PQ	PQ+SPS	SPS
NF- κ B (ngg ⁻¹ tissue)	13.38 $\pm$ 1.16 ^a	85.76 $\pm$ 2.07 ^c	28.44 $\pm$ 1.70 ^b	12.84 $\pm$ 0.92 ^a
TNF- α (ngg ⁻¹ tissue)	5.63 $\pm$ 0.40 ^a	35.21 $\pm$ 1.52 ^c	16.22 $\pm$ 1.89 ^b	5.58 $\pm$ 0.41 ^a
IL-1 β (ngg ⁻¹ tissue)	21.38 $\pm$ 1.12 ^a	86.70 $\pm$ 1.55 ^c	45.69 $\pm$ 1.25 ^b	21.09 $\pm$ 1.20 ^a
IL-6 (ngg ⁻¹ tissue)	4.80 $\pm$ 0.42 ^a	52.70 $\pm$ 1.87 ^c	11.55 $\pm$ 0.65 ^b	4.71 $\pm$ 0.44 ^a
COX-2 (ngg ⁻¹ tissue)	20.92 $\pm$ 0.92 ^a	85.47 $\pm$ 1.36 ^c	36.37 $\pm$ 1.66 ^b	20.04 $\pm$ 0.66 ^a

Table 3. Mean $\pm$ SE of inflammatory indices in control, PQ administrated, co-administrated and SPS administrated groups. Values with dissimilar letters are notably ($P < 0.05$) dissimilar from other groups. PQ: Paraquat; SPS Sciadopitysin; IL-1 β : Interleukin-1 β ; TNF- α : Tumor necrosis factor alpha; IL-6: Interleukin-6; NF- κ B: Nuclear factor-kappa B; COX-2: cyclooxygenase-2.

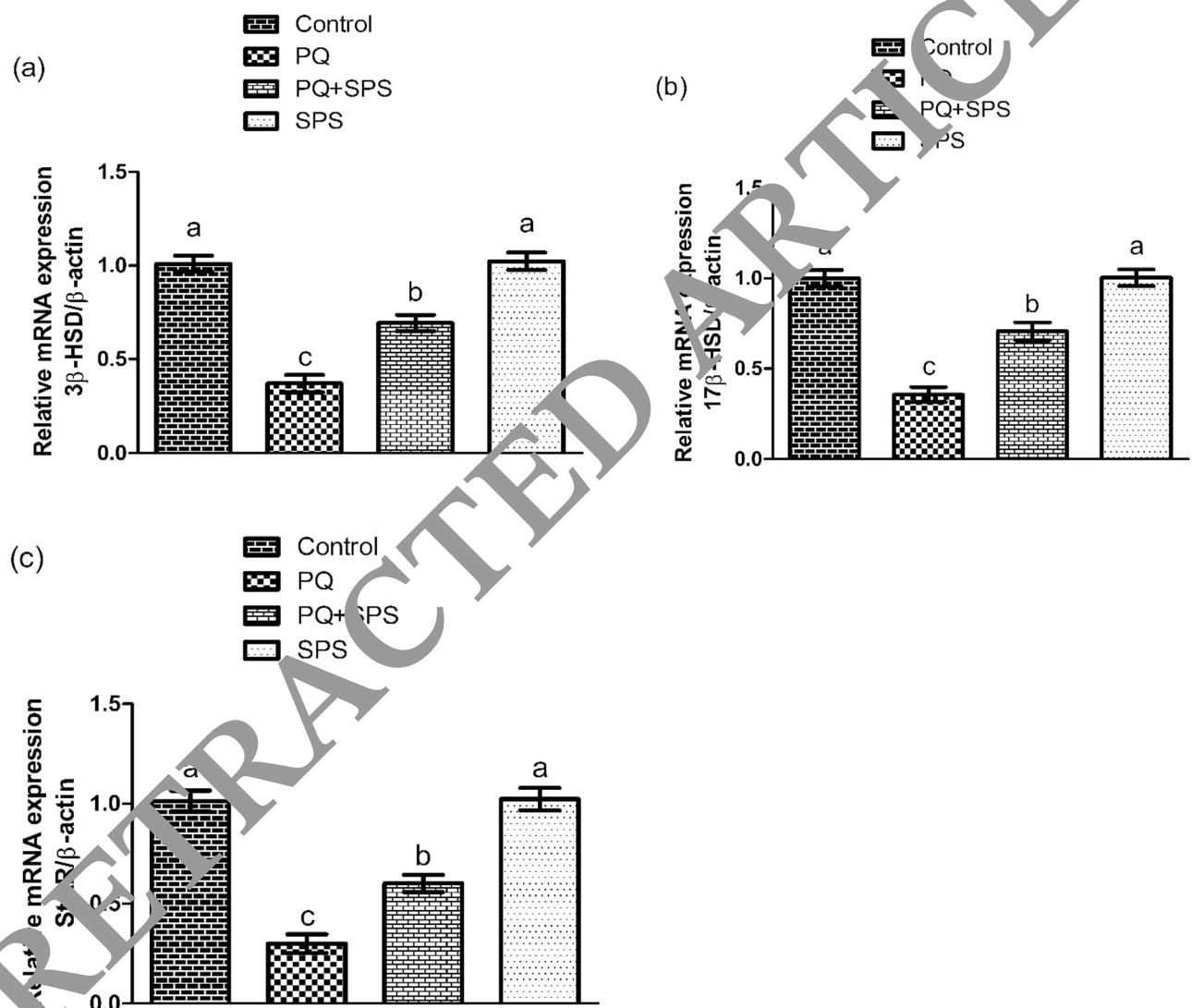


Figure 2. Effect of PQ and SPS on the expression of (a) 3 β -HSD, (b) 17 β -HSD and (c) StAR. Vertical bars are rooted on mean $\pm$ SE values. Different superscripts on bars are showing significant ($P < 0.05$) difference. PQ: Paraquat; SPS Sciadopitysin; 3 β -HSD: 3 β -hydroxysteroid dehydrogenase; 17 β -HSD: 17 β -hydroxysteroid dehydrogenase; StAR: Steroidogenic acute regulatory protein.

Effect of PQ and SPS on histopathology

The exposure to PQ significantly ($P < 0.05$) decreased the epithelial height, the diameter of seminiferous tubules and tunica-propria height. Whereas, a significant ($P < 0.05$) increase was noticed in the tubular lumen and interstitial spaces. Moreover, PQ intoxication significantly ($P < 0.05$) decreased the count of germ cells. However, the co-administration of SPS with PQ significantly ($P < 0.05$) restored these detrimental changes as well as germ cells

Parameters	Groups			
	Control	PQ	PQ + SPS	SPS
FSH (ng/ml)	4.11 ± 0.13 ^a	1.69 ± 0.11 ^c	3.55 ± 0.17 ^b	4.15 ± 0.13 ^a
LH (ng/ml)	3.29 ± 0.13 ^a	1.42 ± 0.12 ^c	2.72 ± 0.11 ^b	3.32 ± 0.12 ^a
Plasma testosterone (ng/ml)	5.29 ± 0.19 ^a	2.52 ± 0.13 ^c	4.28 ± 0.11 ^b	5.33 ± 0.20 ^a

Table 4. Mean ± SE of hormonal axis in control, PQ administrated, co-administrated and SPS administrated groups. Values with dissimilar letters are notably ($P < 0.05$) dissimilar from other groups. PQ: Paraquat; SPS: Sciadopitysin; LH: Luteinizing hormone; FSH: Follicle stimulating hormone and plasma testosterone.

number in co-treated rats in contrast to PQ-administered rats. Furthermore, non-significant alterations were observed between only SPS supplemented and control groups (Fig. 3, Table 5).

Discussion

PQ is a herbicide that is primarily used in the agricultural fields to eliminate weeds¹⁶. PQ causes oxidative damage and cell death through ROS generation i.e., hydrogen peroxide, hydroxyl radical as well as superoxide anion. PQ exposure affects the liver, kidneys, brain, lungs and eventually causes death due to organ failure¹⁷. According to mounting evidences PQ has adverse effects on male reproductive tract due to OS¹⁸. It is reported that PQ exposure increases lipid peroxidation (LP) in the testicular tissues, indicating that oxidative damage may be the reason of deteriorated semen quality, moreover PQ also induces apoptosis in germ cells¹⁹. SPS is an important naturally occurring flavone that displays multiple therapeutic properties²⁰. It has been reported that SPS is an efficient anti-oxidant²¹ that efficiently scavenges free radicals to reduce LP. SPS also shows anti-viral, as anti-cancer, anti-bacterial as well anti-inflammatory properties²². Therefore, the present investigation was designed to evaluate the attenuative role of SPS against PQ-induced testicular toxicity.

The administration of PQ significantly reduced the activities of anti-oxidant enzymes (CAT, GPx, SOD and GSR), on the other hand augmented the level of ROS and MDA²³. Testicular tissues contain a sophisticated system of free radical scavengers i.e., antioxidant enzymes that protect from the adverse effects of OS. Excessive production of free radicals results in disturbance in the antioxidant defense system of male reproductive tract that causes severe damage to the reproductive system²⁴. SOD is a cellular antioxidant enzyme that catalyses the transfer of superoxide into hydrogen peroxide (H_2O_2)²⁵. On the other hand, CAT and GPx convert H_2O_2 to water (H_2O)²⁶. GSR maintains GSH level that supports to maintain the continuous action of GPx²⁷. The increased MDA level is a marker of LP and OS, which is caused by excessive ROS production, leading to disruption of the activities of antioxidant enzymes²⁸. Nevertheless, the supplementation of SPS significantly increased anti-oxidant enzymes activities, on the other hand decreased the levels of ROS as well as MDA, which might be attributed to its potential anti-oxidant potential. The anti-oxidant property of flavonoids is due to the occurrence of multiple OH- groups is broadly attributed to their electron-donating nature, which allow them to neutralize ROS and alleviate OS²⁹.

The administration of PQ significantly reduced the sperm viability, motility and count, whereas escalated the sperm structural damages. According to Deepananda et al.⁸, PQ has the ability to induce spermatogenic failure, as evinced by decreased sperm number along with sperm motility. Due to high content of PUFAs and the absence of cytoplasmic defense in their cell membranes, spermatozoa are highly vulnerable to the harmful effects of ROS³⁰. As documented earlier, OS is responsible for deteriorated sperm quality²⁷. The midpiece section of spermatozoa is particularly highly susceptible to harmful effects of ROS³¹. In comparison to sperm tail and head, the mid-piece segment of spermatozoa is enriched with PUFAs, making it more vulnerable to ROS attack. OS also decreases the production by directly damaging sperms mitochondria, which reduces the function of flagella and results in the immobility of sperms³². Numerous studies have demonstrated that ROS adversely affects the sperm motility, morphology as well as sperm number³³. Nonetheless, the supplementation of SPS significantly reduced all the spermatogenic damages, owing to its free radical scavenging nature.

Apoptosis is a cell death mechanism that is responsible for the eradication of abnormal cells³⁴. In the current investigation, we assessed the expression of Caspase-3, Bcl-2 and Bax. Our findings revealed that PQ exposure up-regulated Bax and Caspase-3 expressions, whereas decreased the Bcl-2 expression. Bcl-2 and Bax both are members of Bcl-2 family. Bcl-2 acts as an anti-apoptotic protein and promotes cell's survival by inhibiting the release of cytochrome c, while Bax acts as an apoptotic protein and prompts apoptosis by increasing the discharge of cytochrome c into the cytoplasm^{35,36}, which plays a crucial role in Caspase-3 activation that eventually results in cell death or apoptosis. However, the supplementation of SPS lowered the Bax and Caspase-3 expressions, on the other hand increased the Bcl-2 expression by regulating Bcl-2/Bax ratio that may be attributed to its anti-apoptotic nature.

In this study, PQ exposure significantly augmented the expression of NF-κB, which induced a significant increase in inflammatory indices i.e., TNF-α, IL-1β, IL-6 and COX-2 activity. According to Rehman et al.³⁷, NF-κB stimulation has a pivotal role in pro-inflammatory cytokines (IL-6, TNF-α, IL-1β and COX-2) expressions that indicates acute inflammation as well as other diseases associated with augmented ROS level. NF-κB stimulation encourages the production of IL-6, TNF-α, IL-1β via gene up-regulation that results in testicular damage³⁸. COX-2 is also an important marker of inflammation that has potential role in inflammation³⁹. However, PQ + SPS co-treatment notably lowered the level of inflammatory markers due to its anti-inflammatory effect. It is previously reported that SPS inhibits receptor activator of nuclear factor-kappa B ligand-induced osteoclastogenesis and bone loss by inhibiting inflammation¹⁵.

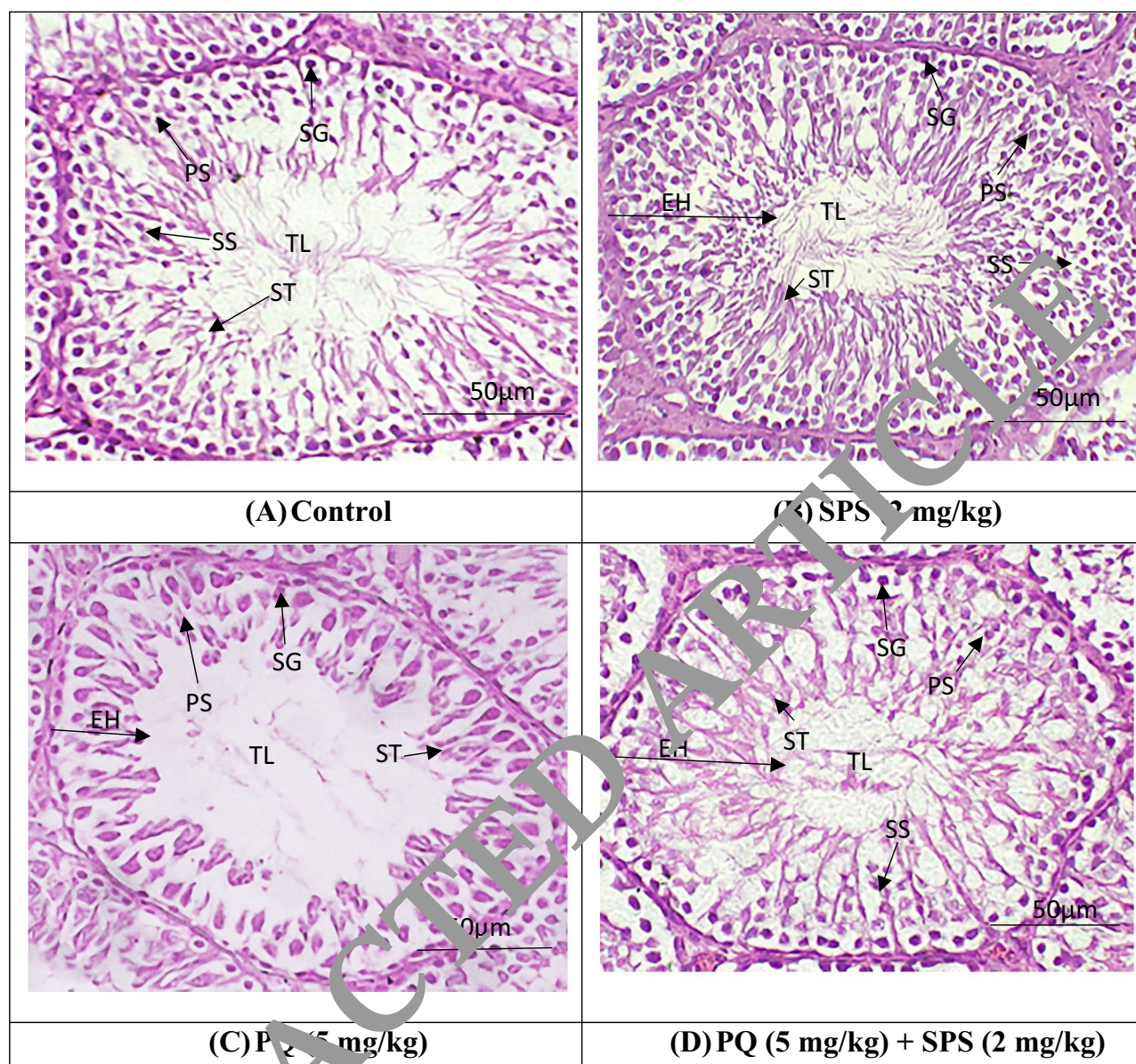


Figure 3. Micro-photographs of the mature albino rat testicles (H&E, 400X): (A) Control group presenting normal seminiferous tubules having small IS and luminal part occupied with germ cells and demonstrating recovered spermatogenesis. (B) SPS administrated group displaying sloughing of the epithelial layer, vacant lumen and deteriorated area of IS; (C) PQ administrated group displaying sloughing of the epithelial layer, vacant lumen and deteriorated area of IS; (D) PQ + SPS co-administrated group representing recovered epithelial part as well as TL having ST and improved the deteriorated IS. PQ: Paraquat; SPS: Sciadopitysin; TL: Tubular lumen; IS: Interstitial spaces; SG: Spermatogonia; PS: Primary spermatocyte; SS: Secondary spermatocyte; ST: Spermatids; EH: Epithelial height.

PQ administration prompted a decrease in 17β -HSD, StAR and 3β -HSD expressions. Both 17β -HSD and 3β -HSD have an important function in testes⁴⁰. StAR is a rate limiting protein and it controls the movement of cholesterol inside the mitochondria for the production of testosterone^{41,42}. The conversion of cholesterol to testosterone occurs through a series of reactions catalyzed by 3β -HSD and 17β -HSD⁴³. Testicular steroidogenesis is a crucial process for producing testosterone and it is undoubtedly mediated by 3β -HSD, 17β -HSD and StAR^{44,45}. Reduced testosterone levels were caused by reduction in the expression of these androgenic enzymes as a result of OS induced by PQ intoxication. However, SPS supplementation prevented the dysregulation in testosterone level by escalating the steroidogenic enzymes expressions that may be attributed to its androgenic property.

The current investigation revealed that exposure to PQ downregulated the level of follicle stimulating hormone (FSH), luteinizing hormone (LH) and plasma testosterone. According to O'Shaughnessy⁴⁶, androgens produced in response to LH and FSH have a critical function in the development and maintenance of sperms. FSH regulates sperm maturation as well as indirectly mediates testicular function. LH stimulates Leydig cells to release

Parameters	Groups			
	Control	PQ	PQ + SPS	SPS
Interstitial spaces (μm)	9.99 $\pm$ 0.88 ^a	68.70 $\pm$ 2.05 ^c	22.47 $\pm$ 1.61 ^b	9.93 $\pm$ 0.90 ^a
Tunica propria (μm)	58.52 $\pm$ 1.94 ^a	13.30 $\pm$ 1.21 ^c	25.84 $\pm$ 1.55 ^b	60.15 $\pm$ 2.25 ^a
Seminiferous tubules (μm)	341.86 $\pm$ 9.65 ^a	113.77 $\pm$ 7.40 ^c	273.86 $\pm$ 9.13 ^b	355.34 $\pm$ 6.40 ^a
Seminiferous tubule epithelial height (μm)	89.87 $\pm$ 3.26 ^a	24.99 $\pm$ 1.57 ^c	75.55 $\pm$ 1.45 ^b	90.59 $\pm$ 2.98 ^a
Tubular lumen (μm)	23.05 $\pm$ 1.66 ^a	85.08 $\pm$ 1.51 ^c	38.53 $\pm$ 1.98 ^b	22.27 $\pm$ 1.51 ^a
Spermatogonia (n)	59.52 $\pm$ 1.52 ^a	23.55 $\pm$ 1.46 ^c	51.41 $\pm$ 1.12 ^b	61.37 $\pm$ 1.21 ^a
Primary spermatocytes (n)	52.29 $\pm$ 1.38 ^a	15.40 $\pm$ 1.27 ^c	36.31 $\pm$ 1.68 ^b	53.17 $\pm$ 1.44 ^a
Secondary spermatocytes (n)	46.55 $\pm$ 1.59 ^a	12.96 $\pm$ 1.22 ^c	38.55 $\pm$ 2.12 ^b	47.98 $\pm$ 1.62 ^a
Spermatids (n)	53.83 $\pm$ 0.96 ^a	24.07 $\pm$ 0.80 ^c	43.77 $\pm$ 1.57 ^b	54.99 $\pm$ 0.88 ^a

Table 5. Mean $\pm$ SE of histopathology in control, PQ administrated, co-administrated and SPS administrated groups. Values with dissimilar letters are notably ($P < 0.05$) dissimilar from other groups. PQ: Paraquat; SPS: Sciadopitysin.

testosterone⁴⁷. Testosterone is a key male reproductive hormone due to its role in regulating spermatogenesis⁴⁸. Therefore, appropriate proportion of LH, FSH as well as testosterone is necessary for spermatogenesis⁴⁹. Reduced LH is unable to stimulate LCs to produce sufficient concentration of testosterone⁵⁰. According to previous studies a decrease in testosterone concentration reduces the number of sperm⁵¹. However, the supplementation of SPS restored the hormonal level that eventually recovered spermatogenesis, possibly by regulating hypothalamic-pituitary-testicles axis.

According to current study, the administration of PQ resulted in severe histopathological damage in testicles such as, reduction in seminiferous tubular diameter, in addition to height and tunica propria width. Besides, tubular lumen diameter and interstitial spaces were decreased. Our findings are also confirmed by the research of Mustafa et al.², who stated that PQ administration induces testicular histopathological damage. Additionally, PQ exposure decreased the germ cells count. According to previous literature, the reduced level of testosterone induces histopathological abnormalities that eventually results in spermatogenic failure⁵². It is also reported that the exposure to PQ induces necrosis of testicular tissue due to excessive ROS generation^{17,53}. Nonetheless, the supplementation of SPS recovered all the impairments owing to its androgenic, anti-oxidant, anti-apoptotic and anti-inflammatory potentials. According to our previous research SPS has the potential to attenuate PQ induced renal damage by lowering the ROS⁵⁴.

Conclusion

In the present investigation, PQ intoxication induced OS by elevating the level of ROS in the testicular tissues and disturbed biochemical, spermatogenic, hormonal along with histopathological profile. Moreover, the intoxication of PQ lowered steroidogenic enzymes and anti-apoptotic markers expressions, while increasing the expressions of apoptotic markers. Additionally, PQ also induced histopathological impairments in the testicular tissues of rats. However, the supplementation of SPS showed protective effects against PQ prompted testicular damage that may be attributed to its androgenic, anti-oxidant, anti-apoptotic and anti-inflammatory nature.

Materials and methods

Chemicals

PQ (CAS No: 75365-73-0, Molecular Weight: 257.16, purity: 98%, Physical form: Solid) and SPS (CAS No: 521-34-2, Molecular Weight: 580.54, purity $\geq$ 95%, Physical form: Solid) were obtained from Sigma-Aldrich, Germany.

Animals

48 adult male albino rats having weight 180–220g (6–8 weeks old) were used in this experiment, animals were housed in steel enclosures at 12 h day/night cycle, temperature (22–25 °C) and humidity (45 $\pm$ 5%). Ad libitum access to food as well as tap water (H_2O) was granted. The ethics council of the University of Agriculture, Faisalabad, provided guidelines for the handling and treatment of animals, which were followed during the whole trial. The study was conducted in accordance with ARRIVE guidelines. The rats were treated and handled according to the protocol of the European Union of Animal Care and Experimentation (CEE Council 86/609) that was further approved by the University of Agriculture institutional ethical committee (DGS/15449-52/19-05-2023).

Experimental layout

Animals ($n = 48$) were randomly distributed into 4 groups ($n = 12$), after one week of acclimatization to the laboratory environment. Group-1 was designated as control, PQ administrated group was exposed to 5 mg/kg of PQ, group-3 (PQ + SPS co-administrated group) received 5 mg/kg of PQ and one hour later 2 mg/kg of SPS, whereas SPS (only) supplemented group was treated with 2 mg/kg of SPS in compliance with the study of El-Aarag et al.⁵⁵ The dose of PQ was selected according to the previous study conducted by Kheiripour et al.⁵⁶. The doses were given through oral gavage. All the treatments were given for 8 weeks as the rats require a duration of about 54–56 days for the completion of one spermatogenic cycle. On the last day (56th) of the experiment, all the rats were anesthetized by using xylazine (6 mg/kg) and ketamine (60 mg/kg) and decapitated⁵⁷, both testicles were

excised, weighed and the right testis was stored at -80°C for biochemical analysis and testes were homogenized in Na_3PO_4 buffer at 12,000 rpm for 15 min at 4°C . This supernatant was finally used to assess various parameters. Whereas, left testis was fixed in 10% formalin solution for histological examination.

Biochemical assessment

The activity of CAT and SOD was determined by following the practice of Afsar et al.⁵⁸. The activity of GPx was measured by using the technique of Lawrence and Burk⁵⁹. GSR content was estimated by using the method of Factor et al.⁶⁰. MDA level was determined by using previously described Razak et al.⁶¹. While the level of ROS was measured by using the protocol of Hayashi et al.⁶².

Semen analysis

Epididymal sperm number was counted using a hemocytometer according to the protocol demonstrated by Yokoi et al.⁶³. Caudal epididymis was excised and crushed with anatomical cutters in 5 mL of physiological saline solution followed by incubation at 37°C for 30 min. The filtrate was diluted (1:100) using a solution containing 5 g of sodium bicarbonate (NaHCO_3), 1 mL of formalin (35%) as well as 25 mg eosin in 100 mL distilled water. A 10 mL droplet of the above solution was put in a sperm counting chamber and examined (10 fields) using a light microscope at 400X. Phase-contrast microscope (400X) was used to record sperm motility and results were expressed in %⁶⁴. Eosin-nigrosin stains were used to assess sperm viability and observed microscopically. Moreover, the sperm morphological abnormalities were evaluated in percentage following the methodology of Filler⁶⁵.

Hypo-osmotic swelling (HOS) test

Sperm integrity was estimated by HOS test in compliance with the methodology of Correa and Zavos⁶⁶. Initially, 20 μL semen was mixed with 180 μL of fructose solution at 80 mOsm/L osmotic pressure for 20 min. Following successive processing as well as incubation, the spermatozoa were stained using eosin-nigrosin. Finally, 200 sperms having swollen and non-swollen tails were estimated using a light microscope (400X).

Real-time Polymerase Chain Reaction (RT-qPCR)

The expressions of steroidogenic enzymes and apoptotic markers were determined by using RT-qPCR. The tissue samples were homogenized using a rotor–stator homogenizer in order to extract RNA. After centrifugation, the supernatant was collected and subsequently mixed with 70% ethanol. RNA spin columns were used for purification. DNase I treatment was performed followed by multiple washes. Ethanol was removed and the extracted RNA was stored at -70°C . RNA quality was determined with the help of NanoDrop Spectrophotometer, using RNA samples having A260/A280 ratio between 1.8 and 2.0. cDNA was synthesized from RNA samples using the Thermo Scientific Revert Aid RT Kit and reverse transcriptase enzyme. RT-qPCR was carried out in a LightCycler 480 instrument (Roche, Basel, Switzerland) using Syber green PCR master mix (TaKaRa, Biotech. Co., Ltd.). PCR amplification was initiated by denaturation for 3 min at 95°C that was followed by 45 cycles of 95°C for 10 s, 60°C for 40 s, 72°C for 32 s and a final incubation at 75°C for 5 min. The analyses were performed by using the $2^{-\Delta\Delta\text{CT}}$ by considering β -actin as an internal control⁶⁷. The primer sequences of target genes are displayed in Table 6 as reported previously.

Gene	Primers 5'→3'	Accession number
β -HSD	Forward: GCATCCTGAAAAATGGTGGC Reverse: GCCACATTGCCTACATACAC	NM_001007719
17 β -HSD	Forward: CAGCTTCCAAGGCTTTTGTG Reverse: CAGGTTTCAGCTCCAATCGT	NM_054007
StAR	Forward: AAAAGGCCTTGGGCATACTC Reverse: CATAGAGTCTGTCCATGGGC	NM_031558
Bax	Forward: GGCCTTTTGTGCTACAGGGTT Reverse: AGCTCCATGTTGTTGTCCAG	NM_017059.2
Bcl-2	Forward: ACAACATCGCTCTGTGGAT Reverse: TCAGAGACAGCCAGGAGAA	NM_016993.1
Caspase-3	Forward: ATCCATGGAAGCAAGTCGAT Reverse: CCTTTTGCTGTGATCTTCCT	NM_012922.2
β -actin	Forward: TACAGCTTCACCACCACAGC Reverse: GGAACCGCTCATTGCCGATA	NM_031144

Table 6. Primers sequences for the real-time quantitative reverse transcription-polymerase (RT-qPCR). β -HSD: β -hydroxysteroid dehydrogenase; 17 β -HSD: 17 β -hydroxysteroid dehydrogenase; StAR: Steroidogenic acute regulatory protein; Bcl-2: B-cell lymphoma 2; Bax: Bcl-2-associated X protein; Caspase-3: Cysteine–aspartic acid protease.

Inflammatory markers analysis

The Level of NF- κ B (CSB-E13148r), TNF- α (CSB-E07379r), IL-1 β (CSB-E08055r), IL-6 (CSB-E04640r) and COX-2 (CSB-E13399r) activity were estimated by ELISA kits (Cusabio Technology LLC, Houston, TX, USA) following the manufacturer's guidelines. Firstly, 50 μ L of sample was dispensed to the microplate wells. After that, 50 μ L of antibody cocktail was poured to the wells. Plates were incubated at room temperature for the duration of 1 h. After washing properly with the help of wash buffer, 100 μ L of TMB substrates were dispensed to each well and incubated for about 10 min. After the addition of 100 μ L of stop solution, the color was developed. The optical density was noted at 450 nm using Tecan Multimode Reader.

Hormonal assay

The levels of FSH (serial number-H101), LH (serial number-H206) and plasma testosterone (serial number-H090) were assessed by enzyme linked immune-sorbent assay (ELISA) kits (Los Angeles, CA USA) in accordance with the manufacturer's guidelines. 50 μ L of assay diluent and 10 μ L of plasma were added to 96-well ELISA plate and incubation was performed for about 2 h at room temperature. Then, plates were rinsed with the deionized water and before adding 100 μ L of peroxidase-conjugated immunoglobulin G (IgG) and FSH solution, anti-LH or anti-testosterone in each well, incubation was carried out for maximum 2 h. Plates were again rinsed with the deionized water, substrate solution was added in wells and incubated for about 25 min at room temperature. 50 μ L of stop solution was added into each well to terminate the reaction. Finally, the absorbance of FSH, LH and plasma testosterone was recorded at 450 nm. All samples were run in triplicates and performed at a same time under same conditions to avoid inter-assay variation.

Histopathology

For the purpose of histopathological assessment, testicular tissues were fixed in 10% formalin buffer solution and dehydrated in rising concentration of ethyl alcohol. After that, tissues were embedded in paraffin wax. 5 μ m thick slices were cut and stained using H & E which were later observed under a light microscope. Image-J2X program was used to analyze the photomicrographs of the slides.

Statistical analysis

Data were shown as Mean $\pm$ SE. The normal distribution of the data was checked by Shapiro–Wilk test, whereas the homogeneity of variances was checked and confirmed by using Levene test. Using one-way ANOVA and Tukey's test, data were statistically examined through Minitab software. Significance level was set at $P < 0.05$.

Ethical approval

The rats were treated and handled according to the protocol of the European Union of Animal Care and Experimentation (CEE Council 86/609) that was further approved by the University of Agriculture institutional ethical committee (DGS/15449-52/19-03-2023).

Data availability

All the data is contained in the manuscript.

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Author contributions

M.U.I. and M.Q. conceived and planned the study. A.H., A.A. and T.A. accomplished the experiment and histology, and helped in compiling the results. M.A., A.A. and S.R. assisted in statistical analysis and writing the results. M.U.I., M.Q. and A.H. wrote the manuscript. All the authors read and approved the final manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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